Recursive Programs

Bird & de Moor, *Algebra of Programming* ch. 6 — in Freyd diagram order

Conventions

Composition is JUXTAPOSITION in diagram order: $RS =$ "first R, then S". Objects α, β, γ; relations Q, R, S, T, U, V, X, Y; **maps** (entire and simple) f, g, h, k, m; coreflexives $C \subset 1$. R° converse, $\subset$ containment, $\cap, \cup$ meet/join, $-$ subtraction, $\emptyset$ empty, Π largest, 1 identity, $\mathrm{dom}, \mathrm{ran}$. $[\alpha]$ power object, $\ni : [\alpha] \to \alpha$ membership, Λ the power transpose ($f = \Lambda R \iff f\ni = R$), $\tau = \Lambda 1$ singleton, $\bigcup = E\ni$ union, $ER = \Lambda(\ni R)$ existential image. F, G relators; $\mathsf{in} : F(\mu F) \to \mu F$ the initial algebra, $(\!|R|\!)$ its catamorphism; $(\mu X : \varphi X)$ the least solution of $X = \varphi X$. Points: $x \, R \, y$.

B&dM dictionary. Everything else on the sheet is the book's, mirrored through this table and nothing more.

B&dM	here	
$S \cdot R$	RS	composition reverses
id	1	
$\mathrm{outl}, \mathrm{outr}$	π, π'	
$\mathrm{inl}, \mathrm{inr}$	ι, ι'	
α	in	the initial algebra, not an object
$(\lvert R \rvert)$	$(\!\lvert R \rvert\!)$	
μ	$\bigcup$	(μ is kept for the least fixed point)
$\in, \ \ni$	$\ni, \ \ni^\circ$	
$T \setminus U, \ U/T$	$U/T, \ T \setminus U$	slashes flip, roles kept
$(a, b) \, \langle R, S \rangle \, c$	$c \, \langle R, S \rangle \, (a, b)$	points reverse too

A row tagged ✓ has its mirrored statement machine-checked in `AOP/A*.lean`; the rest are hand-mirrored from the page.

6.1 Digits of a number

Convert a positive natural to its decimal digits. The pattern of the whole chapter in one example: specify a function as a refinement of some relation, discover that the relation solves a recursion equation, then read the recursion off as a program.

6.1 Decimal ::= wrap Digit^+ | snoc(Decimal, Digit), so ([wrap, snoc], Decimal) is the initial algebra of $FA = \mathrm{Digit}^+ + (A \times \mathrm{Digit})$.

$\mathrm{val} = (\!|\mathrm{embed}, \mathrm{op}|\!) : \mathrm{Decimal} \to \mathrm{Nat}^+$, with embed the inclusion of digits and $\mathrm{op}(n, d) = 10n + d$.

(6.1) $\mathrm{digits} \subset \mathrm{val}^\circ$ — a *functional refinement* of val°, which is not yet known to be a map. This is the specification.

$$
\begin{aligned}
& \mathrm{val}^\circ \\
= \quad & \{\text{definition}\} \\
& (\!|\mathrm{embed}, \mathrm{op}|\!)^\circ \\
= \quad & \{\text{catamorphisms}\} \\
& ([\mathrm{wrap}, \mathrm{snoc}]^\circ (F \ \mathrm{val})[\mathrm{embed}, \mathrm{op}])^\circ \\
= \quad & \{\text{converse}\} \\
& [\mathrm{embed}, \mathrm{op}]^\circ (F \mathrm{val}^\circ)[\mathrm{wrap}, \mathrm{snoc}] \\
= \quad & \{\text{definition of } F\} \\
& [\mathrm{embed}, \mathrm{op}]^\circ (1 + (\mathrm{val}^\circ \times 1))[\mathrm{wrap}, \mathrm{snoc}] \\
= \quad & \{\text{coproduct}\} \\
& [\mathrm{embed}, \mathrm{op}]^\circ [\mathrm{wrap}, (\mathrm{val}^\circ \times 1) \ \mathrm{snoc}] \\
= \quad & \{\text{coproduct}\} \\
& (\mathrm{embed}^\circ \ \mathrm{wrap}) \cup (\mathrm{op}^\circ (\mathrm{val}^\circ \times 1) \ \mathrm{snoc})
\end{aligned}
$$

6.1 ✓ So $\mathrm{val}^\circ = (\mathrm{embed}^\circ \ \mathrm{wrap}) \cup (\mathrm{op}^\circ (\mathrm{val}^\circ \times 1) \ \mathrm{snoc})$.

$\mathrm{op}(n, d) = m \iff n = m \ \mathrm{div} \ 10 \wedge d = m \ \mathrm{mod} \ 10$, so op° is defined exactly for $m \geq 10$ and embed° exactly for $m < 10$: the two branches have disjoint domains, and the join becomes a conditional.

$\mathrm{val}^\circ m = \mathrm{wrap} \ m$ if $m < 10$, else $\mathrm{snoc}(\mathrm{val}^\circ (m \ \mathrm{div} \ 10), m \ \mathrm{mod} \ 10)$. The recursion terminates because $m \geq 10 \implies m \ \mathrm{div} \ 10 < m$, so val° is a total function and $\mathrm{digits} = \mathrm{val}^\circ$.

$\mathrm{digits} \ m = [m]$ if $m < 10$, else $\mathrm{digits}(m \ \mathrm{div} \ 10) \mathbin{+\!\!+} [m \ \mathrm{mod} \ 10]$ — quadratic, because snoc on lists is linear.

With an accumulator, $\mathrm{digits} \ m = f(m, [])$ where $f(m, x) = [m] \mathbin{+\!\!+} x$ if $m < 10$, else $f(m \ \mathrm{div} \ 10, [m \ \mathrm{mod} \ 10] \mathbin{+\!\!+} x)$ — linear, and correct at 0.

6.2 Least fixed points

T 6.1 ✓ **KNASTER–TARSKI** *φ a monotonic mapping on the arrows $\alpha \to \beta$ of a locally complete allegory. Then $\varphi X \subset X$ and $\varphi X = X$ have the same least solution, written $(\mu X : \varphi X)$; dually $X \subset \varphi X$ and $X = \varphi X$ have the same greatest solution.*

Proof: $(\mu X : \varphi X) = \bigcap\{X : \varphi X \subset X\}$. Local completeness is what makes that meet exist.

in is an isomorphism, so $(\!|R|\!)$ is the unique — hence both least and greatest — solution of $X = \mathsf{in}^\circ(FX)R$, and $\varphi X = \mathsf{in}^\circ(FX)R$ is monotonic. Knaster–Tarski then gives:

(6.2) ✓ $(\!|R|\!) \subset X \Longleftarrow \mathsf{in}^\circ(FX)R \subset X$.

(6.3) ✓ $X \subset (\!|R|\!) \Longleftarrow X \subset \mathsf{in}^\circ(FX)R$.

(6.4) ✓ **FUSION, $\subset$** $(\!|T|\!) \subset (\!|R|\!)S \Longleftarrow (FS)T \subset RS$.

(6.5) ✓ **FUSION, $\supset$** $(\!|R|\!)S \subset (\!|T|\!) \Longleftarrow RS \subset (FS)T$. Equality of the two hypotheses gives back the catamorphism fusion law of ch. 3.

Ex 6.3 ✓ **KLEENE** φ *continuous* — it preserves joins of ascending chains — then $(\mu X : \varphi X) = \bigcup\{\varphi^n\emptyset : 0 \leq n\}$.

Ex 6.4 ✓ **FIXED POINT INDUCTION** $(\mu X : \varphi X) \subset A \Longleftarrow \varphi A \subset A$; by Kleene, $(\forall X.\, X \subset A \Longrightarrow \varphi X \subset A)$ suffices too.

Ex 6.5 The least solution of $\varphi X \subset X$ satisfies $\varphi X = X$ — Lambek's lemma, for the containment order read as a category and φ as a functor on it.

Ex 6.7 ✓ $(\!|R|\!) \subset (\!|S|\!) \Longleftarrow (F(\!|S|\!))R \subset (F(\!|S|\!))S$. It does *not* need $R \subset S$.

Ex 6.8 ✓ R **DIFUNCTIONAL** $\triangleq R = RR^\circ R$; the difunctional closure of R is a least fixed point.

6.3 Hylomorphisms

6.3 A **HYLOMORPHISM** is a catamorphism after the converse of one, $(\!|S|\!)^\circ(\!|R|\!)$ for $R : F\alpha \to \alpha$ and $S : F\beta \to \beta$. The initial type μF is the *intermediate data structure*: it is built by $(\!|S|\!)^\circ$ and consumed by $(\!|R|\!)$, and never appears in the answer.

T 6.2 ✓ **HYLOMORPHISM THEOREM**

$$(\!|S|\!)^\circ(\!|R|\!) = (\mu X : S^\circ(FX)R).$$

S° divides, FX conquers, R combines.

Proof in two halves: $(\!|S|\!)^\circ(\!|R|\!)$ is a fixed point (functors and the two catamorphism laws), and it is below every prefixed point — that half goes through division, $(\!|S|\!)^\circ(\!|R|\!) \subset X \Longleftrightarrow (\!|R|\!) \subset X/(\!|S|\!)^\circ$, which is the standard move for least fixed points.

C 6.1 ✓ $FX = GX + HX$, so an F-algebra is a pair:

$$(\!|S_1, S_2|\!)^\circ(\!|R_1, R_2|\!) = (\mu X : (S_1^\circ(GX)R_1) \cup (S_2^\circ(HX)R_2)).$$

in is an isomorphism and $(\!|\mathsf{in}|\!) = 1$, so every catamorphism and every converse of one is already a hylomorphism.

Ex 6.10 ✓ Hylomorphisms preserve simplicity: R and S simple $\Longrightarrow (\!|S|\!)^\circ(\!|R|\!)$ simple.

6.4 Fast exponentiation and modulus

6.4 exp $a = (\!|\text{one}, \text{mult } a|\!) : \text{Nat} \to \text{Nat}$ encodes $a^0 = 1$ and $a^{b+1} = a \times a^b$ — $O(b)$ steps. Divide and conquer brings it to $O(\log b)$.

Bin = listl Bit, Bit $= \{0, 1\}$; convert $= (\!|\text{zero}, \text{shift}|\!) : \text{Bin} \to \text{Nat}$ with $\text{shift}(n, d) = 2n + d$ reads a bit sequence as a number. It is simple.

$$
\begin{array}{ll}
 & \text{exp } a \\
\supseteq & \quad \{\text{convert simple, so convert}^\circ \text{ convert} \subset 1\} \\
 & \text{convert}^\circ \text{ convert } (\text{exp } a) \\
= & \quad \{\text{fusion, with op } a(n,d) = (d = 0 \to n^2, a \times n^2)\} \\
 & \text{convert}^\circ (\!|\text{one}, \text{op } a|\!) \\
= & \quad \{\text{Corollary 6.1}\} \\
 & (\mu X : (\text{zero}^\circ \text{ one}) \cup (\text{shift}^\circ (X \times 1)(\text{op } a)))
\end{array}
$$

6.4 ✓ The fusion step needs zero (exp a) = one and shift (exp a) = (exp $a \times 1$)(op a). zero and shift have disjoint ranges, so the join is again a conditional:

exp $a\,b = 1$ if $b = 0$, else op $a(\text{exp } a(b \text{ div } 2), b \text{ mod } 2)$.

6.4 ✓ Same idea for a mod b: mod $b = (\!|\text{zero}, \text{succ } b|\!)$ with succ $b\,a = (a = b - 1 \to 0, a + 1)$ costs $O(a)$; fusing through convert gives

$$\text{mod } b \supset (\mu X : (\text{zero}^\circ \text{ zero}) \cup (\text{shift}^\circ (X \times 1)(\text{op } b)))$$

with op $b(r, d) = (n \geq b \to n - b, n)$ and $n = 2r + d$.

Conditions: zero (mod b) = zero and shift (mod b) = (mod $b \times 1$)(op b). The program is mod $b\,a = 0$ if $a = 0$; op $b(\text{mod } b(a \text{ div } 2), 0)$ if a even; op $b(\text{mod } b(a \text{ div } 2), 1)$ if a odd — $O(\log a)$.

6.5 Unique fixed points

A hylomorphism is the *least* fixed point of its recursion, not necessarily the only one.

6.5 On Nat take $X = (\!|\text{zero}, \text{positive}|\!)^\circ (\!|\text{zero}, 1|\!)$ with positive = succ° succ the coreflexive on positives. Both catamorphisms describe the coreflexive on 0, so X is that. But its recursion $X = (\text{zero}^\circ \text{ zero}) \cup (\text{positive } X)$ also has $X = 1$ as a solution.

$[\text{zero}, \text{positive}]^\circ$ and $[\text{zero}, 1]$ are both maps, yet the least solution is not even entire. Simplicity does carry over (Ex 6.10); entireness is what needs a hypothesis, and the hypothesis is that a membership relation be *inductive*.

Inductive and well-founded relations

6.5 ✓ $R : \alpha \to \alpha$ is **INDUCTIVE** $\triangleq$ for all X,

$$X/R \subset X \implies \Pi \subset X.$$

Pointwise: if $(\forall c.\, cRa \implies cXb) \implies aXb$ holds for all a, b, then X holds everywhere — exactly a proof by induction over R.

$<$ on Nat gives ordinary mathematical induction; tail = $\text{cons}^\circ \pi'$ gives induction over lists.

Ex 6.13 ✓ S inductive and $RR \subset SR \implies R$ inductive. Hence $R \subset S$ with S inductive $\implies R$ inductive, and S inductive $\iff S^+$ inductive, where $S^+ = (\mu X : S \cup (SX))$.

Ex 6.15 ✓ The meet of two inductive relations is inductive; the join need not be.

6.5 ✓ R is **WELL-FOUNDED** $\triangleq$ for all X, $X \subset RX \implies X = \emptyset$ — no infinite chain $a_0 R a_1 R \ldots$ Inductive $\implies$ well-founded, and the converse holds in a Boolean allegory.

Ex 6.16 ✓ R well-founded $\implies fRf^\circ$ well-founded, for any map f.

Membership

6.5 A relator F may carry a **MEMBERSHIP** arrow $\text{mem}(F)_\alpha : F\alpha \to \alpha$ with $a \text{ mem } x$ exactly when a is an element of x. For the polynomial relators, the power relator and the type relators it exists:

$\text{mem}(\text{Id}) = 1$, $\text{mem}(K_\beta) = \emptyset$, $\text{mem}(F + G) = [\text{mem}(F), \text{mem}(G)]$.

$\text{mem}(F \times G) = (\pi \text{ mem}(F)) \cup (\pi' \text{ mem}(G))$, $\text{mem}(F \circ G) = \text{mem}(F) \text{ mem}(G)$, $\text{mem}([\cdot]) = \ni$.

Type relators, via sets: $\text{mem}(T) = \text{setify}(T) \ni$ with $\text{setify}(T) = \Lambda(\text{mem}(T))$ — Ex 6.17 gives a set-free alternative.

mem is a lax natural transformation $1 \hookrightarrow F$, that is $(FR) \text{ mem} \subset \text{mem } R$, and it is the *largest* one of that type — so a membership relation, if it exists, is unique.

6.5 ✓ The result the section is for: $\text{in}^\circ \text{ mem}(F)$ is inductive. It is the immediate-subterm relation.

Nat, $FX = 1 + X$: mem $= [\emptyset, 1]$ and $[\text{zero}, \text{succ}]^\circ [\emptyset, 1] = \text{succ}^\circ$ is inductive; $< = \text{pred}^+$ with pred $= \text{succ}^\circ$, which is what made §6.1's recursion terminate.

Lists: mem $= [\emptyset, \pi']$ and $[\text{nil}, \text{cons}]^\circ [\emptyset, \pi'] = \text{cons}^\circ \pi' =$ tail is inductive, so proper suffix tail^+ is; on snoc-lists init and proper prefix are.

Unique solutions

T 6.3 ✓ *$S\,\mathrm{mem}(F)$ inductive $\implies X = S(FX)R$ has a unique solution $\varphi(R,S)$, and $\varphi(R,S)$ is entire if R and S are.* With $S := S^\circ$ this is the hylomorphism of §6.3, now pinned down as the *only* fixed point.

C 6.2 ✓ $g\,\mathrm{mem}(F)$ inductive $\implies$ the unique solution of $X = g(FX)f$ is a map ($X = (\!|g^\circ|\!)^\circ (\!|f|\!)$ is entire by the theorem and simple by Ex 6.10).

C 6.3 ✓ $R^\circ\ \mathrm{mem}(F)$ inductive $\implies (\!|R|\!)$ is surjective when R is (R surjective $\triangleq 1 \subset R^\circ R$, i.e. $\mathrm{ran}\ R = 1$).

T 6.4 ✓ *R surjective and $Rf \subset (Ff)$* in $\implies f^\circ = (\!|R|\!)$. This is the theorem §5.6 borrowed to show partition $=$ concat$^\circ$ is a catamorphism.

Ex 6.12 ✓ R inductive $\iff X = X/R$ has a unique solution.

Ex 6.17 $\mathrm{inlist} : \mathrm{list}^+\alpha \to \alpha$ is a catamorphism, but inlist on list α is not; instead $\mathrm{inlist} = \mathrm{tail}^*\ \mathrm{head}$. The same trick defines intree.

Ex 6.18 The formal definition: mem is a membership relation for F if $(1/\ \mathrm{mem})(FR) = R/\ \mathrm{mem}$ for all R; equivalently $(S/\ \mathrm{mem})(FR) = (SR)/\ \mathrm{mem}$ for all R, S.

Ex 6.20 $\mathrm{mem}(G)/\ \mathrm{mem}(F)$ is the largest lax natural transformation $F \hookrightarrow G$.

Ex 6.21 In a Boolean allegory, $\mathrm{mem}(F)$ is entire $\iff F\emptyset = \emptyset$.

6.6 Sorting by selection

(6.6) ✓ $\mathrm{sort} \subset \mathrm{perm}\ (\mathrm{ordered})$, for R a *connected* preorder ($R \cup R^\circ = \Pi$). A connected preorder need not be anti-symmetric, so sort is genuinely a refinement, not an equality.

$\mathrm{ordered} = (\!|\mathrm{nil}, \mathrm{ok}\,\mathrm{cons}|\!)$, where the coreflexive ok holds at (a,x) when $(\forall b : b\,\mathrm{inlist}\,x : aRb)$ — rebuild the list, checking at each step that only R-smaller elements go in front. ($\mathrm{ok}(a,x) = (x = [] \vee aR(\mathrm{head}\ x))$ also works but is less useful here.)

Selection sort

$$
\begin{aligned}
& \mathrm{perm}\ (\mathrm{ordered}) \\
= \ & \quad \text{perm = perm}^\circ \text{ and ordered = ordered}^\circ \\
& (\mathrm{ordered}\ \mathrm{perm})^\circ \\
= \ & \quad \text{ordered} = (\!|\text{nil, ok cons}|\!) \\
& ((\!|\mathrm{nil}, \mathrm{ok}\,\mathrm{cons}|\!)\ \mathrm{perm})^\circ \\
\supseteq \ & \quad \text{fusion, for a suitable select} \\
& (\!|\mathrm{nil}, \mathrm{select}^\circ|\!)^\circ
\end{aligned}
$$

6.6 The fusion proviso is $\mathrm{ok}\,\mathrm{cons}\,\mathrm{perm} \supset (1 \times \mathrm{perm})\mathrm{select}^\circ$, and select is found by calculating:

$$
\begin{aligned}
& \mathrm{ok}\,\mathrm{cons}\,\mathrm{perm} \\
= \ & \quad \text{perm} = (\!|\text{nil, cons perm}|\!)\ \ (\S 5.6) \\
& \mathrm{ok}\ (1 \times \mathrm{perm})\ \mathrm{cons}\,\mathrm{perm} \\
= \ & \quad (1 \times \text{perm})\text{ok} = \text{ok}(1 \times \text{perm}),\ \text{Ex 6.22} \\
& (1 \times \mathrm{perm})\ \mathrm{ok}\,\mathrm{cons}\,\mathrm{perm} \\
\supseteq \ & \quad \text{specifying select} \subseteq \text{perm cons}^\circ\ \text{ok} \\
& (1 \times \mathrm{perm})\mathrm{select}^\circ
\end{aligned}
$$

6.6 In words: $(a,y) = \mathrm{select}\ x$ means $[a] \mathbin{+\!\!+} y$ is a permutation of x with aRb for every b in y. It is undefined on $[]$, so take $\mathrm{select} = \mathrm{embed}\ (\!|\mathrm{base}, \mathrm{step}|\!)$ with $\mathrm{embed} : \mathrm{list}\ \alpha \to \mathrm{list}^+\alpha$.

Fusion conditions: $\mathrm{base} \subset \mathrm{wrap}\,\mathrm{perm}\,\mathrm{cons}^\circ\ \mathrm{ok}$ and $(1 \times (\mathrm{cons}^\circ\ \mathrm{ok}))\ \mathrm{step} \subset \mathrm{cons}\,\mathrm{perm}\,\mathrm{cons}^\circ\ \mathrm{ok}$, satisfied by $\mathrm{base}\ a = (a, [])$ and $\mathrm{step}(a,(b,x)) = (a, [b] \mathbin{+\!\!+} x)$ if aRb, else $(b, [a] \mathbin{+\!\!+} x)$.

6.6 ✓ The hylomorphism theorem then makes $X = (\!|\mathrm{nil}, \mathrm{select}^\circ|\!)^\circ$ the unique solution of $X = (\mathrm{nil}^\circ\ \mathrm{nil}) \cup (\mathrm{select}\ (1 \times X)\ \mathrm{cons})$, i.e. $\mathrm{sort}\ x = []$ if $x = []$, else $[a] \mathbin{+\!\!+} \mathrm{sort}\ y$ where $(a,y) = \mathrm{select}\ x$.

Quicksort

6.6 Same shape, with a tree as the intermediate datatype: $\mathrm{tree}\ \alpha ::= \mathrm{null} \mid \mathrm{fork}(\mathrm{tree}\ \alpha, \alpha, \mathrm{tree}\ \alpha)$ and $\mathrm{flatten} = (\!|\mathrm{nil}, \mathrm{join}|\!)$, $\mathrm{join}(x,a,y) = x \mathbin{+\!\!+} [a] \mathbin{+\!\!+} y$.

$$
\begin{aligned}
& \mathrm{perm}\ (\mathrm{ordered}) \\
\supseteq \ & \quad \text{flatten a map, so flatten}^\circ\ \text{flatten} \subset 1 \\
& \mathrm{perm}\,\mathrm{flatten}^\circ\ \mathrm{flatten}\,\mathrm{ordered} \\
= \ & \quad \text{claim: flatten ordered = inordered flatten} \\
& \mathrm{perm}\,\mathrm{flatten}^\circ\ \mathrm{inordered}\,\mathrm{flatten} \\
= \ & \quad \text{converses} \\
& (\mathrm{inordered}\,\mathrm{flatten}\,\mathrm{perm})^\circ\ \mathrm{flatten} \\
\supseteq \ & \quad \text{fusion, for a suitable split} \\
& (\!|\mathrm{nil}, \mathrm{split}^\circ|\!)^\circ\ \ \mathrm{flatten}
\end{aligned}
$$

6.6 $\mathrm{inordered} = (\!|\mathrm{null}, \mathrm{check}\,\mathrm{fork}|\!)$, with check holding at (x,a,y) when $(\forall b : b\,\mathrm{intree}\,x \implies bRa)$ and $(\forall b : b\,\mathrm{intree}\,y \implies aRb)$; check$'$ is the same with inlist for intree. Writing $Ff = f \times 1 \times f$, the proviso is $F(\mathrm{flatten}\,\mathrm{perm})\mathrm{split}^\circ \subset \mathrm{check}\,\mathrm{fork}\,\mathrm{flatten}\,\mathrm{perm}$.

Discharged by taking $\mathrm{split} \subset \mathrm{perm}\,\mathrm{join}^\circ\mathrm{check}'$, i.e. $(y, a, z) = \mathrm{split}\ x$ means $y +\!\!+ [a] +\!\!+ z$ is a permutation of x with bRa throughout y and aRb throughout z.

$\mathrm{split} = \mathrm{embed}\ (\!|\mathrm{base}, \mathrm{step}|\!)$ with $\mathrm{base} \subset \mathrm{wrap}\,\mathrm{perm}\,\mathrm{join}^\circ\mathrm{check}'$ and $(1 \times (\mathrm{join}^\circ\mathrm{check}'))\ \mathrm{step} \subset \mathrm{cons}\,\mathrm{perm}\,\mathrm{join}^\circ\mathrm{check}'$ (the book prints split for step here, and drops the $^\circ$ on join), satisfied by $\mathrm{base}\ a = ([], a, [])$ and $\mathrm{step}(a, (x, b, y)) = ([a] +\!\!+ x, b, y)$ if aRb, else $(x, b, [a] +\!\!+ y)$.

6.6 $X = (\!|\mathrm{nil}, \mathrm{split}^\circ|\!)^\circ$ flatten is the least solution of $X = (\mathrm{nil}^\circ\ \mathrm{nil}) \cup (\mathrm{split}(X \times 1 \times X)\mathrm{join})$, i.e. $\mathrm{sort}\ x = []$ if $x = []$, else $\mathrm{sort}\ y +\!\!+ [a] +\!\!+ \mathrm{sort}\ z$ where $(y, a, z) = \mathrm{split}\ x$.

Ex 6.22 $\mathrm{inlist} = \mathrm{perm}\,\mathrm{inlist}$; hence a point-free ok and $(1 \times \mathrm{perm})\mathrm{ok} = \mathrm{ok}(1 \times \mathrm{perm})$.

Ex 6.24 $\mathrm{ordered}(R)\,\mathrm{ordered}(S) = \mathrm{ordered}(R \cap S)$.

Ex 6.25 Sorting a *set*: $\mathrm{sort} \subset \mathrm{setify}^\circ\ \mathrm{ordered}(R)$ is the wrong specification, $\mathrm{sort} \subset \mathrm{setify}^\circ\ \mathrm{ordered}(R \cap \mathrm{neq})$ the right one, where $a\,\mathrm{neq}\,b$ iff $a \neq b$.

Ex 6.30 From $\mathrm{perm} = (\!|\mathrm{nil}, \mathrm{add}|\!)$ instead, fusion gives $\mathrm{perm}(\mathrm{ordered}) = (\!|\mathrm{nil}, \mathrm{add}\,\mathrm{ordered}|\!) \supset (\!|\mathrm{nil}, \mathrm{insert}|\!)$ for a suitable map insert with $(1 \times \mathrm{ordered})\mathrm{insert} \subset \mathrm{add}\,\mathrm{ordered}$ — insertion sort.

6.7 Closure

6.7 ✓ R^*, the **REFLEXIVE TRANSITIVE CLOSURE**, is the smallest preorder containing R:

$$R \subset X \iff R^* \subset X \quad \text{for every preorder } X.$$

(6.7) ✓ $R^* = (\mu X : 1 \cup (RX))$.

(6.8) ✓ $R^* = (\mu X : 1 \cup (XR))$. That $S = (\mu X : 1 \cup (RX))$ is reflexive and contains R is immediate; transitivity goes through division, the same move as Theorem 6.2.

Ex 6.32 ✓ $SR^* = (\mu X : S \cup (XR))$, $RS^* = (\mu X : R \cup (XS))$, $S^*R = (\mu X : R \cup (SX))$.

Ex 6.33 $R^* = (\!|[1, 1]|\!)^\circ (\!|[1, R]|\!)$, the intermediate datatype being $\mathrm{iterate}\ \alpha ::= \mathrm{once}\ \alpha \mid \mathrm{again}(\mathrm{iterate}\ \alpha)$.

Ex 6.34 $R^* = \mathrm{chainR}^\circ\ \mathrm{head}$ for a catamorphism chainR on non-empty lists.

Ex 6.36 $(R \cup S)^* = (R^*S)^*R^*$, by the diagonal rule.

Ex 6.37 $CR = C \implies R^* = (\sim\!C\ R)^*$, for C coreflexive.

When the recursion has a unique solution

6.7 $X = 1 \cup (RX)$ has a unique solution — necessarily R^* — exactly when R is inductive. tail is inductive, so $\mathrm{suffix} = 1 \cup (\mathrm{tail}\,\mathrm{suffix})$ pins suffix down, and transposing gives $\Lambda\ \mathrm{suffix} = \langle \tau, \Lambda(\mathrm{tail}\,\mathrm{suffix})\rangle\ \mathrm{cup}$.

tail is undefined on $[]$, so a case analysis splits it: $(\Lambda\ \mathrm{suffix})[] = \{[]\}$ and $(\Lambda\ \mathrm{suffix})([a] +\!\!+ x) = \{[a] +\!\!+ x\} \cup (\Lambda\ \mathrm{suffix})x$ — sets as lists, and this is §5.6's $\mathrm{tails}\ [] = [[]]$, $\mathrm{tails}([a] +\!\!+ x) = [[a] +\!\!+ x] +\!\!+ \mathrm{tails}\ x$.

Computing a closure that is not a unique fixed point

6.7 **SUBTRACTION** (Ex 4.30) $R - S \subset T \iff R \subset S \cup T$; hence $R - \emptyset = R$, $R \cup S = R \cup (S - R)$, $R - (S \cup T) = R - S - T$ and $(R \cup S) - T = (R - T) \cup (S - T)$.

Ex 6.35 ✓ **M-CALCULUS** the two defining rules are $\varphi(\mu X : \varphi X) = (\mu X : \varphi X)$ and $\varphi Y \subset Y \implies (\mu X : \varphi X) \subset Y$. From them: *rolling* $(\mu X : \varphi(\psi X)) = \varphi(\mu X : \psi(\varphi X))$, *diagonal* $(\mu X : \mu Y : \varphi(X, Y)) = (\mu X : \varphi(X, X))$, *substitution* $(\mu X : \varphi(\mu Y : \psi(X, Y))) = (\mu X : \psi(\varphi X, X))$.

(6.9) $\theta(P, Q) \triangleq P \cup (\mu X : Q \cup ((XR) - P))$, which satisfies

$$\theta(\emptyset, S) = SR^*, \quad \theta(P, \emptyset) = P,$$

$$\theta(P, Q) = \theta(P \cup Q, (QR) - P - Q).$$

The third comes out of the rolling rule and the subtraction laws.

Read as a program: compute $\theta(\emptyset, S)$, where $\theta(P, Q) = P$ if $Q = \emptyset$, else $\theta(P \cup Q, (QR) - P - Q)$. It terminates only if SR^* is finite; then P grows at every step.

Relations as data are awkward, so apply Λ throughout: with $p = \Lambda P$, $q = \Lambda Q$, $s = \Lambda S$ and $\Lambda(SR^*) = (\Lambda S)(ER^*)$,

$$\mathrm{close}(p, q) = \begin{cases} p & \text{if } q = \emptyset \\ \mathrm{close}(p \cup q, (ER)q - p - q) & \text{otherwise,} \end{cases}$$

now with set union and set difference. This is how $E(R^*)$ is computed whenever the answer is known to be finite — reachability in a graph.